Maintain SolidFire eSDS

Element Software

NetApp
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Maintain SolidFire eSDS

Find information about using maintenance mode and replacing the drives in your SolidFire eSDS clusters.

- Get the license key
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- Links to troubleshooting KB articles
- Contents of the sf_sds_config.yaml file

Find more information

- NetApp SolidFire Resources Page
- Documentation for earlier versions of NetApp SolidFire and Element products

Get the license key

You should get the license key information for your SolidFire eSDS cluster before you contact NetApp Support for help with troubleshooting issues. The license key for the SolidFire eSDS cluster consists of the cluster's order number combined with the serial number.

You can use the `GetLicenseKey` method to get the license key information for the SolidFire eSDS storage cluster. The `GetLicenseKey` method has no input parameters. This method has the following return values:

Name	Description	Type
`orderNumber`	The new sales order number of the storage cluster.	string
`serialNumber`	The new serial number of the storage cluster.	string

Step

1. Run the `GetLicenseKey` API method as shown in the following example:

   Requests for this method are similar to the following example:

```
{
   "method": "GetLicenseKey",
   "params": {
   },
   "id": 1
}
```

This method returns a response similar to the following example:

```
{
   "id" : 1,
   "result" : {
      "serialNumber": "30G56E3WV",
      "orderNumber": "33601"
     }
   }
}
```

Find more information

- NetApp SolidFire Resources Page
- Documentation for earlier versions of NetApp SolidFire and Element products

Use maintenance mode on SolidFire eSDS clusters

If you need to take a storage node offline for maintenance such as software upgrades or host repairs, you can minimize the I/O impact to the rest of the storage cluster by enabling maintenance mode for that node.

If you want to verify the current status of maintenance mode on your node, use the `ListActiveNodes` API method. The node object includes a `maintenanceMode` parameter, which indicates the current status of maintenance mode on the node.

Ensure that you do the maintenance as soon as maintenance mode is enabled. Do not leave the node in maintenance mode any more than necessary.

You can transition a storage node to maintenance mode only if the node is healthy (has no blocking cluster faults) and the storage cluster is tolerant to a single node failure. After you enable maintenance mode for a healthy and tolerant node, the node is not immediately transitioned; it is monitored until the following conditions are true:

- All volumes hosted on the node have failed over and the node is no longer hosting as the primary for any volume.
- A temporary standby node is assigned for every volume being failed over.

After these criteria are met, the node is transitioned to maintenance mode. If these criteria are not met within a five-minute period, the node will not enter maintenance mode.

When you disable maintenance mode for a storage node, the node is monitored until the following conditions are true:

- All data is fully replicated to the node.
- All blocking cluster faults are resolved.
- All temporary standby node assignments for the volumes hosted on the node have been inactivated.

After these criteria are met, the node is transitioned out of maintenance mode. If these criteria are not met within one hour, the node will fail to transition out of maintenance mode.

Possible scenarios while using maintenance mode

- If a node is in maintenance mode, but has not been rebooted yet, and/or has not had maintenance occur, or it had maintenance occur and is back up and healthy, but you have not disabled maintenance mode, and another node goes down, maintenance mode on the first node will be disabled automatically.
- If one of your nodes is in maintenance mode, and another node goes down at the same time, there will be an outage. You have to wait till the node that is in maintenance mode comes back online.
- If you put a node that is a member of an ensemble in maintenance mode for a long period of time, the system will automatically remove it from the ensemble, if there are other nodes available to be added in its place.

Enable maintenance mode

You can enable maintenance mode using the `EnableMaintenanceMode` API method. This method has the following input parameters:

Name	Description	Type	Default value	Required
`forceWithUnresolvedFaults`	Force maintenance mode to be enabled for this node even with blocking cluster faults present.	boolean	False	No
`nodes`	The list of node IDs to put in maintenance mode. Only one node at a time is supported.	integer array	None	Yes
`perMinutePrimarySwapLimit`	The number of primary slices to swap per minute. If not specified, all primary slices will be swapped at once.	integer	None	No

Name	Description	Type	Default value	Required
`timeout`	Specifies how long maintenance mode should remain enabled before it is automatically disabled. Formatted as a time string (for example, HH:mm:ss). If not specified, maintenance mode will remain enabled until explicitly disabled.	string	None	No

This method has the following return values:

Name	Description	Type
`asyncHandle`	You can use the `GetAsyncResult` method to retrieve this asyncHandle and determine when the maintenance mode transition is complete.	integer
`currentMode`	The current maintenance mode state of the node. Possible values: • Disabled: No maintenance has been requested. • FailedToRecover: The node failed to recover from maintenance mode. • RecoveringFromMaintenance: The node is in the process of recovering from maintenance mode. • PreparingForMaintenance: Actions are being taken to prepare a node to have maintenance performed. • ReadyForMaintenance: The node is ready for maintenance to be performed.	MaintenanceMode (string)

Name	Description	Type
`requestedMode`	The requested maintenance mode state of the node. Possible values: • Disabled: No maintenance has been requested. • FailedToRecover: The node failed to recover from maintenance mode. • RecoveringFromMaintenance: The node is in the process of recovering from maintenance mode. • PreparingForMaintenance: Actions are being taken to prepare a node to have maintenance performed. • ReadyForMaintenance: The node is ready for maintenance to be performed.	MaintenanceMode (string)

Disable maintenance mode

You can disable maintenance mode using the `DisableMaintenanceMode` API method. This method has the following input parameter:

Name	Description	Type	Default value	Required
`nodes`	List of storage node IDs to take out of maintenance mode.	integer array	None	Yes

This method has the following return values:

Name	Description	Type
`asyncHandle`	You can use the `GetAsyncResult` method to retrieve this asyncHandle and determine when the maintenance mode transition is complete.	integer

Name	Description	Type
`currentMode`	The current maintenance mode state of the node. Possible values: • Disabled: No maintenance has been requested. • FailedToRecover: The node failed to recover from maintenance mode. • Unexpected: The node was found to be offline, but was in the Disabled mode. • RecoveringFromMaintenance: The node is in the process of recovering from maintenance mode. • PreparingForMaintenance: Actions are being taken to prepare a node to have maintenance performed. • ReadyForMaintenance: The node is ready for maintenance to be performed.	MaintenanceMode (string)
`requestedMode`	The requested maintenance mode state of the node. Possible values: • Disabled: No maintenance has been requested. • FailedToRecover: The node failed to recover from maintenance mode. • Unexpected: The node was found to be offline, but was in the Disabled mode. • RecoveringFromMaintenance: The node is in the process of recovering from maintenance mode. • PreparingForMaintenance: Actions are being taken to prepare a node to have maintenance performed. • ReadyForMaintenance: The node is ready for maintenance to be performed.	MaintenanceMode (string)

Find more information

- NetApp SolidFire Resources Page
- Documentation for earlier versions of NetApp SolidFire and Element products

Replace drives for HPE DL360

Choose from the procedures listed here to replace a drive proactively, replace a drive after it has failed, and replace a cache drive. replace a metadata drive or a block drive in your SolidFire eSDS cluster. The Element UI **Cluster > Drives** page shows the drive wear information.

- Replace a drive proactively
- Replace a faulty drive
- Replace a cache drive

Replace a drive proactively

Perform this procedure if you want to proactively replace a metadata drive or a block drive in your SolidFire eSDS cluster. The Element UI **Cluster** > **Drives** page shows the drive wear information.

What you'll need

- From the NetApp Element software UI, ensure that your cluster is in good health and there are no warnings or cluster faults. You can access the Element UI by using the management virtual IP (MVIP) address of the primary cluster node.
- Ensure that there are no active jobs running on the cluster.
- Ensure that you have familiarized yourself with all the steps.
- Ensure that you take necessary precautions to prevent electrostatic discharge (ESD) while handling drives.

Steps

1. Perform the following steps in the Element UI:
   a. In the Element UI, select **Cluster** > **Drives** > **Active**.
   b. Select the drive that you want to replace.
   c. Make a note of the serial number of the drive. This will help you locate the corresponding BayID in the IPMI interface of the node (HPE Integrated Lights-Out or iLO, in this case).
   d. Select **Bulk Actions** > **Remove**. After you remove the drive, the drive goes into the **Removing** state. It stays in the **Removing** state for a while, waiting for the data on the drive to be synced or redistributed to the remaining drives in the cluster. After the remove is complete, the drive moves to the **Available** state.
2. Perform the following steps to locate the drive slot of the drive that you are replacing:
   a. Log in to the IPMI interface of the node (iLO in this case).
   b. Select **System Information** from the left-hand navigation, and then select **Storage**.
   c. Match the serial number you made a note of in the previous step with what you see on the screen.
   d. Look for the slot number listed against the serial number. This is the physical slot from which you must remove the drive.

3. Now that you have identified the drive, physically remove it as follows:
   a. Identify the drive bay.

      The following image shows the front of the server with the drive bay numbering shown on the left side of the image:

   b. Press the power button on the drive that you want to replace. The LED blinks for 5-10 seconds and stops.
   c. After the LED stops blinking and the drive is powered off, remove it from the server by pressing the red button and pulling the latch.

      > Ensure that you handle drives very carefully.

      After you physically remove the drive, the drive state changes to **Failed** in the Element UI.

4. In the Element UI, select **Cluster** > **Drives** > **Failed**.
5. Select the icon under **Actions**, and then select **Remove**.

   Now you can go ahead and install the new drive in the node.

6. Make a note of the serial number of the new drive.
7. Insert the replacement drive by carefully pushing the drive into the bay using the latch and closing the latch. The drive powers on when inserted correctly.
8. Perform the following steps to verify the new drive details in iLO:
   a. Log in to iLO.
   b. Select **Information** > **Integrated Management Log**. You will see an event logged for the drive that you added.
   c. Select **System Information** from the left-hand navigation, and then select **Storage**.
   d. Scroll till you find information about the bay that you replaced the drive in.
   e. Verify that the serial number on your screen matches the serial number of the new drive that you replaced.
9. Add the new drive information in the `sf_sds_config.yaml` file for the node in which you replaced the drive.

   The `sf_sds_config.yaml` file is stored in `/opt/sf/`. This file includes all the information about the drives in the node. Every time you replace a drive, you must enter the replacement drive information in this file. For more information about this file, see Contents of the sf_sds_config.yaml file.

   a. Establish an SSH connection to the node by using PuTTY.
   b. In the PuTTY configuration window, enter node MIP in the **Host Name (or IP address)** field.
   c. Select **Open**.
   d. In the terminal window that opens, log in with your username and password.
   e. Run the `# cat /opt/sf/sf_sds_config.yaml` command to list the contents of the file.

f. Replace the entries in the `dataDevices` or `cacheDevices` lists for the drive you replaced with the new drive information.

g. Run `# systemctl start solidfire-update-drives`.

   You see the Bash prompt after this command runs. You should go to the Element UI after this to add the drive to the cluster. The Element UI shows an alert for a new drive that is available.

10. Select **Cluster** > **Drives** > **Available**.

    You see the serial number of the new drive that you installed.

11. Select the icon under **Actions**, and then select **Add**.
12. Refresh the Element UI after the block sync job completes. You see that the alert about the drive available has cleared if you access the **Running Tasks** page from the **Reporting** tab of the Element UI.

Replace a faulty drive

If your SolidFire eSDS cluster has a faulty drive, the Element UI displays an alert. Before you remove the drive from the cluster, verify the reason for failure by looking at the information in the IPMI interface for your node/server. These steps are applicable if you are replacing a block drive or a metadata drive.

What you'll need

- From the NetApp Element software UI, verify that the drive has failed. Element displays an alert when a drive fails. You can access the Element UI by using the management virtual IP (MVIP) address of the primary cluster node.
- Ensure that you have familiarized yourself with all the steps.
- Ensure that you take necessary precautions to prevent electrostatic discharge (ESD) while handling drives.

Steps

1. Remove the failed drive from the cluster as follows using the Element UI:
   a. Select **Cluster** > **Drives** > **Failed**.
   b. Note the node name and serial number associated with the failed drive.
   c. Select the icon under **Actions**, and then select **Remove**. If you see warnings of the service associated with the drive, wait until bin sync completes, and then remove the drive.
2. Perform the following steps to verify the drive failure and view the events logged that are associated with the drive failure:
   a. Log in to the IPMI interface of the node (iLO in this case).
   b. Select **Information** > **Integrated Management Log**. The reason for the drive failure (for example, SSDWearOut) and the location is listed here. You can also see an event stating that the status of the drive is degraded.
   c. Select **System Information** from the left-hand navigation, and then select **Storage**.
   d. Verify the information available about the failed drive. The status of the failed drive will say **Degraded**.
3. Remove the drive physically as follows:
   a. Identify the drive slot number in the chassis.

      The following image shows the front of the server with the drive bay numbering shown on the left side of the image:

b. Press the power button on the drive that you want to replace. The LED blinks for 5-10 seconds and stops.

c. After the LED stops blinking and the drive is powered off, remove it from the server by pressing the red button and pulling the latch.

> Ensure that you handle drives very carefully.

4. Insert the replacement drive by carefully pushing the drive into the bay using the latch and closing the latch. The drive powers on when inserted correctly.
5. Verify the new drive details in iLO:
   a. Select **Information** > **Integrated Management Log**. You see an event logged for the drive that you added.
   b. Refresh the page to see the events logged for the new drive that you added.
6. Verify the health of your storage system in iLO:
   a. Select **System Information** from the left-hand navigation, and then select **Storage**.
   b. Scroll till you find information about the bay in which you installed the new drive.
   c. Make a note of the serial number.
7. Add the new drive information in the `sf_sds_config.yaml` file for the node in which you replaced the drive.

   The `sf_sds_config.yaml` file is stored in `/opt/sf/`. This file includes all the information about the drives in the node. Every time you replace a drive, you must enter the replacement drive information in this file. For more information about this file, see Contents of the sf_sds_config.yaml file.

   a. Establish an SSH connection to the node by using PuTTY.
   b. In the PuTTY configuration window, enter node MIP in the **Host Name (or IP address)** field.
   c. Select **Open**.
   d. In the terminal window that opens, log in with your username and password.
   e. Run the `# cat /opt/sf/sf_sds_config.yaml` command to list the contents of the file.
   f. Replace the entries in the `dataDevices` or `cacheDevices` lists for the drive you replaced with the new drive information.
   g. Run `# systemctl start solidfire-update-drives`.

      You see the Bash prompt after this command runs. You should go to the Element UI after this to add the drive to the cluster. The Element UI shows an alert for a new drive that is available.
8. Select **Cluster** > **Drives** > **Available**.

   You see the serial number of the new drive that you installed.
9. Select the icon under **Actions**, and then select **Add**.
10. Refresh the Element UI after the block sync job completes. You see that the alert about the drive available

has cleared if you access the **Running Tasks** page from the **Reporting** tab of the Element UI.

Replace a cache drive

Perform this procedure if you want to replace the cache drive in your SolidFire eSDS cluster. The cache drive is associated with metadata services. The Element UI **Cluster** > **Drives** page shows the drive wear information.

What you'll need

- From the NetApp Element software UI, ensure that your cluster is in good health and there are no warnings or cluster faults. You can access the Element UI by using the management virtual IP (MVIP) address of the primary cluster node.
- Ensure that there are no active jobs running on the cluster.
- Ensure that you have familiarized yourself with all the steps.
- Ensure that you remove metadata services from the Element UI.
- Ensure that you take necessary precautions to prevent electrostatic discharge (ESD) while handling drives.

Steps

1. Perform the following steps in the Element UI:
   a. In the Element UI, select **Cluster** > **Nodes** > **Active**.
   b. Make a note of the node ID and management IP address of the node in which you are replacing the cache drive.
   c. If the cache drive is healthy and you are proactively replacing it, select **Active Drives**, locate the metadata drive, and remove it from the UI.

      After you remove it, the metadata drive goes to **Removing** state first, and then to **Available**.

   d. If you are performing replacement after the cache drive failed, the metadata drive will be in **Available** state, and listed under **Cluster** > **Drives** > **Available**.
   e. In the Element UI, select **Cluster** > **Drives** > **Active**.
   f. Select the metadata drive associated with the NodeName, where you want to do the cache drive replacement.
   g. Select **Bulk Actions** > **Remove**. After you remove the drive, the drive goes into the **Removing** state. It stays in the **Removing** state for a while, waiting for the data on the drive to be synced or redistributed to the remaining drives in the cluster. After the remove is complete, the drive moves to the **Available** state.
2. Perform the following steps to locate the drive slot of the cache drive that you are replacing:
   a. Log in to the IPMI interface of the node (iLO in this case).
   b. Select **System Information** from the left-hand navigation, and then select **Storage**.
   c. Locate the cache drive.

      Cache drives are of lesser capacity than storage drives.

   d. Look for the slot number listed for cache drive. This is the physical slot from which you must remove the drive.
3. Now that you have identified the drive, physically remove it as follows:

a. Identify the drive bay.

   The following image shows the front of the server with the drive bay numbering shown on the left side of the image:

b. Press the power button on the drive that you want to replace. The LED blinks for 5-10 seconds and stops.

c. After the LED stops blinking and the drive is powered off, remove it from the server by pressing the red button and pulling the latch.

   Ensure that you handle drives very carefully.

   After you physically remove the drive, the drive state changes to **Failed** in the Element UI.

4. Make a note of the HPE model number and the ISN (serial number) of the new cache drive.
5. Insert the replacement drive by carefully pushing the drive into the bay using the latch and closing the latch. The drive powers on when inserted correctly.
6. Perform the following steps to verify the new drive details in iLO:

   a. Log in to iLO.

   b. Select **Information** > **Integrated Management Log**. You see an event logged for the drive that you added.

   c. Select **System Information** from the left-hand navigation, and then select **Storage**.

   d. Scroll till you find information about the bay that you replaced the drive in.

   e. Verify that the serial number on your screen matches the serial number of the new drive that you installed.

7. Add the new cache drive information in the `sf_sds_config.yaml` file for the node in which you replaced the drive.

   The `sf_sds_config.yaml` file is stored in `/opt/sf/`. This file includes all the information about the drives in the node. Every time you replace a drive, you should enter the replacement drive information in this file. For more information about this file, see Contents of the sf_sds_config.yaml file.

   a. Establish an SSH connection to the node by using PuTTY.

   b. In the PuTTY configuration window, enter node MIP address (that you made a note of from the Element UI earlier) in the **Host Name (or IP address)** field.

   c. Select **Open**.

   d. In the terminal window that opens, log in with your username and password.

   e. Run the `nvme list` command to list the NMVe devices.

      You can see the model number and serial number of the new cache drive. See the following sample output:

```
[root@NLABICT062226 ~]# nvme list
Node             SN                   Model                                    Namespace Usage
---------------- -------------------- ---------------------------------------- --------- --------------------------
/dev/nvme0n1     KI9AN0136I020AU17    VK003840KWWFP                            1           3.84  TB /   3.84  TB
/dev/nvme10n1    PHKE913200XM375AGN   EO000375KWJUC                            1         375.08  GB / 375.08  GB
/dev/nvme1n1     KI9AN0136I02AAU0J    VK003840KWWFP                            1           3.84  TB /   3.84  TB
/dev/nvme2n1     KI05T0003I1205C14    VK003840KWWFP                            1           3.84  TB /   3.84  TB
/dev/nvme3n1     KI05T0003I1205C0W    VK003840KWWFP                            1           3.84  TB /   3.84  TB
/dev/nvme4n1     KI05T0003I1205C1O    VK003840KWWFP                            1           3.84  TB /   3.84  TB
/dev/nvme5n1     KI05T0003I1205C1P    VK003840KWWFP                            1           3.84  TB /   3.84  TB
/dev/nvme7n1     KI05T0003I1205C1L    VK003840KWWFP                            1           3.84  TB /   3.84  TB
/dev/nvme8n1     KI05T0003I1205C13    VK003840KWWFP                            1           3.84  TB /   3.84  TB
/dev/nvme9n1     KI9AN0136I02AAU1Q    VK003840KWWFP                            1           3.84  TB /   3.84  TB
[root@NLABICT062226 ~]#
```

f. Add the new cache drive information in `/opt/sf/sf_sds_config.yaml`.

   You should replace the existing cache drive model number and serial number with the corresponding information for the new cache drive. See the following example:

```
schemaVersion: "2.0"

network:
  managementInterface: "team1G"
  storageInterface: "team10G"
dataDrives:
  - "/dev/nvme0n1"
  - "/dev/nvme1n1"
  - "/dev/nvme2n1"
  - "/dev/nvme3n1"
  - "/dev/nvme4n1"
  - "/dev/nvme5n1"
  - "/dev/nvme7n1"
  - "/dev/nvme8n1"
  - "/dev/nvme9n1"
cacheDevices:
  - "/dev/disk/by-id/nvme-EO000375KWJUC_PHKE913200XM375AGN"
```

Model number

Serial number

g. Save the `/opt/sf/sf_sds_config.yaml` file.

8. Perform the steps for the scenario that is applicable to you:

Scenario	Steps
The new inserted cache drive shows up after you run the `nvme list` command	1. Run `# systemctl restart solidfire`. This takes around three minutes. 2. Check the `solidfire` status by running `system status solidfire`. 3. Go to step 9.
The new inserted cache drive does not show up after you run the `nvme list` command	1. Reboot the node. 2. After the node reboots, verify that the `solidfire` services are running by logging in to the node (using PuTTY), and running the `system status solidfire` command. 3. Go to step 9.

> Restarting `solidfire` or rebooting the node causes some cluster faults, which eventually clear in about five minutes.

9. In the Element UI, add back the metadata drive that you removed:
   a. Select **Cluster** > **Drives** > **Available**.
   b. Select the icon under Actions, and select **Add**.
10. Refresh your Element UI after the block sync job completes.

    You can see that the alert about the drive available has cleared along with other cluster faults.

Find more information

- NetApp SolidFire Resources Page
- Documentation for earlier versions of NetApp SolidFire and Element products

Replace drives for Dell R640

Choose from the procedures listed here to replace a drive proactively, replace a drive after it has failed, and replace a cache drive. replace a metadata drive or a block drive in your SolidFire eSDS cluster. The Element UI **Cluster > Drives** page shows the drive wear information.

- Replace a drive proactively
- Replace a faulty drive
- Replace a cache drive

Replace a drive proactively

Perform this procedure if you want to proactively replace a metadata drive or a block drive in your SolidFire eSDS cluster. The Element UI **Cluster** > **Drives** page shows the drive wear information.

What you'll need

- From the NetApp Element software UI, ensure that your cluster is in good health and there are no warnings or cluster faults. You can access the Element UI by using the management virtual IP (MVIP) address of the primary cluster node.
- Ensure that there are no active jobs running on the cluster.
- Ensure that you have familiarized yourself with all the steps.
- Ensure that you take necessary precautions to prevent electrostatic discharge (ESD) while handling drives.

Steps

1. Perform the following steps in the Element UI:
   a. In the Element UI, select **Cluster** > **Drives** > **Active**.
   b. Select the drive that you want to replace.
   c. Make a note of the serial number of the drive. This will help you locate the corresponding slot number of the drive in the Integrated Dell Remote Access Controller (iDRAC).

d. Select **Bulk Actions** > **Remove**. After you remove the drive, the drive goes into the **Removing** state. It stays in the **Removing** state for a while, waiting for the data on the drive to be synced or redistributed to the remaining drives in the cluster. After the remove is complete, the drive moves to the **Available** state.

2. Perform the following steps to locate the drive slot of the drive that you are replacing:
   a. Log in to the IPMI interface of the node (iDRAC in this case).
   b. Select **Storage** from the menu, and then select **Physical Disks**.
   c. To find the serial number of the drive, select the **+** icon next to each PCIe SSD.
   d. Match the serial number you see on the screen with what you noted in the Element UI.
   e. Look for the slot number listed against the serial number.

      This is the physical slot from which you should remove the drive.

3. Now that you have identified the drive, physically remove it as follows:
   a. Identify the drive slot number in the chassis.

      The following image shows the front of the server with the slot numbering shown for each drive:

   b. Press the button on the drive.

      The latch comes out.

   c. Physically pull the drive out from the slot.

      Ensure that you handle drives very carefully.

      After you physically remove the drive, the drive state changes to **Failed** in the Element UI.

4. In the Element UI, select **Cluster** > **Drives** > **Failed**.
5. Select the icon under **Actions**, and then select **Remove**.

   Now you can go ahead and install the new drive in the node.

6. Make a note of the serial number of the new drive.
7. Insert the replacement drive by carefully pushing the drive into the bay using the latch and closing the latch. The drive powers on when inserted correctly.
8. Perform the following steps to verify the new drive details in iDRAC:
   a. Log in to iDRAC.
   b. Select **Maintenance** > **System Event Log**.

      You see an event logged for the drive that you added.

   c. Select **Storage** from the menu, and then select **Physical Disks**.
   d. Verify that the new drive you inserted shows up in the corresponding slot in the UI.

e. To find the serial number of the drive, select the **+** icon next to each PCIe SSD.

9. Add the new drive information in the `sf_sds_config.yaml` file for the node in which you replaced the drive.

   The `sf_sds_config.yaml` file is stored in `/opt/sf/`. This file includes all the information about the drives in the node. Every time you replace a drive, you must enter the replacement drive information in this file. For more information about this file, see Contents of the sf_sds_config.yaml file.

   a. Establish an SSH connection to the node by using PuTTY.
   b. In the PuTTY configuration window, enter node MIP in the **Host Name (or IP address)** field.
   c. Select **Open**.
   d. In the terminal window that opens, log in with your username and password.
   e. Run the `# cat /opt/sf/sf_sds_config.yaml` command to list the contents of the file.
   f. Replace the entries in the `dataDevices` or `cacheDevices` lists for the drive you replaced with the new drive information.
   g. Run `# systemctl start solidfire-update-drives`.

      You see the Bash prompt after this command runs. You should go to the Element UI after this to add the drive to the cluster. The Element UI shows an alert for a new drive that is available.

10. Select **Cluster** > **Drives** > **Available**.

    You see the serial number of the new drive that you installed.

11. Select the icon under **Actions**, and then select **Add**.
12. Refresh the Element UI after the block sync job completes. You see that the alert about the drive available has cleared if you access the **Running Tasks** page from the **Reporting** tab of the Element UI.

Replace a faulty drive

If your SolidFire eSDS cluster has a faulty drive, the Element UI displays an alert. Before you remove the drive from the cluster, verify the reason for failure by looking at the information in the IPMI interface for your node/server. These steps are applicable if you are replacing a block drive or a metadata drive.

What you'll need

- From the NetApp Element software UI, verify that the drive has failed. Element displays an alert when a drive fails. You can access the Element UI by using the management virtual IP (MVIP) address of the primary cluster node.
- Ensure that you have familiarized yourself with all the steps.
- Ensure that you take necessary precautions to prevent electrostatic discharge (ESD) while handling drives.

Steps

1. Remove the failed drive from the cluster as follows using the Element UI:
   a. Select **Cluster** > **Drives** > **Failed**.
   b. Note the node name and serial number associated with the failed drive.
   c. Select the icon under **Actions**, and then select **Remove**. If you see warnings of the service associated with the drive, wait until bin sync completes, and then remove the drive.

2. Perform the following steps to verify the drive failure and view the events logged that are associated with the drive failure:
   a. Log in to the IPMI interface of the node (IDRAC in this case).
   b. Select **Maintenance** > **System Event Log** to see the reason for the drive failure (for example, SSDWearOut or drive not inserted properly).

      You can also see an event showing the status of the drive.

   c. Select **Storage** from the menu, and then select **Physical Disks**.
   d. Find the slot number of the failed drive using the serial number that you noted in the Element UI.
3. Remove the drive physically as follows:
   a. Identify the drive slot number in the chassis.

      The following image shows the front of the server with the slot numbering shown for each drive:

   b. Press the button on the drive.

      The latch comes out.

   c. Physically pull the drive out from the slot.

      Ensure that you handle drives very carefully.

4. Insert the replacement drive by carefully pushing the drive into the slot using the latch and closing the latch.

   The drive powers on when inserted correctly.

5. Verify the new drive details in iDRAC:
   a. Select **Maintenance** > **System Event Log**. You see an event logged for the drive that you added.
   b. Select **Storage** from the menu, and then select **Physical Disks**.
   c. Verify that the new drive you inserted shows up in the corresponding slot in the UI.
   d. To find the serial number of the drive, select the **+** icon next to each PCIe SSD.
6. Add the new drive information in the `sf_sds_config.yaml` file for the node in which you replaced the drive.

   The `sf_sds_config.yaml` file is stored in `/opt/sf/`. This file includes all the information about the drives in the node. Every time you replace a drive, you must enter the replacement drive information in this file. For more information about this file, see Contents of the sf_sds_config.yaml file.

   a. Establish an SSH connection to the node by using PuTTY.
   b. In the PuTTY configuration window, enter node MIP in the **Host Name (or IP address)** field.
   c. Select **Open**.
   d. In the terminal window that opens, log in with your username and password.
   e. Run the `# cat /opt/sf/sf_sds_config.yaml` command to list the contents of the file.

f. Replace the entries in the `dataDevices` or `cacheDevices` lists for the drive you replaced with the new drive information.

g. Run `# systemctl start solidfire-update-drives`.

   You see the Bash prompt after this command runs. You should go to the Element UI after this to add the drive to the cluster. The Element UI shows an alert for a new drive that is available.

7. Select **Cluster** > **Drives** > **Available**.

   You see the serial number of the new drive that you installed.

8. Select the icon under **Actions**, and then select **Add**.
9. Refresh the Element UI after the block sync job completes. You see that the alert about the drive available has cleared if you access the **Running Tasks** page from the **Reporting** tab of the Element UI.

Replace a cache drive

Perform this procedure if you want to replace the cache drive in your SolidFire eSDS cluster. The cache drive is associated with metadata services. The Element UI **Cluster** > **Drives** page shows the drive wear information.

What you'll need

- From the NetApp Element software UI, ensure that your cluster is in good health and there are no warnings or cluster faults. You can access the Element UI by using the management virtual IP (MVIP) address of the primary cluster node.
- Ensure that there are no active jobs running on the cluster.
- Ensure that you have familiarized yourself with all the steps.
- Ensure that you remove metadata services from the Element UI.
- Ensure that you take necessary precautions to prevent electrostatic discharge (ESD) while handling drives.

Steps

1. Perform the following steps in the Element UI:

   a. In the Element UI, select **Cluster** > **Nodes** > **Active**.

   b. Make a note of the node ID and management IP address of the node in which you are replacing the cache drive.

   c. If the cache drive is healthy and you are proactively replacing it, select **Active Drives**, locate the metadata drive, and remove it from the UI.

      After you remove it, the metadata drive goes to **Removing** state first, and then to **Available**.

   d. If you are performing replacement after the cache drive failed, the metadata drive will be in **Available** state, and listed under **Cluster** > **Drives** > **Available**.

   e. In the Element UI, select **Cluster** > **Drives** > **Active**.

   f. Select the metadata drive associated with the NodeName, where you want to do the cache drive replacement.

   g. Select **Bulk Actions** > **Remove**. After you remove the drive, the drive goes into the **Removing** state. It stays in the **Removing** state for a while, waiting for the data on the drive to be synced or redistributed to the remaining drives in the cluster. After the remove is complete, the drive moves to the **Available**

state.

2. Perform the following steps to locate the drive slot of the cache drive that you are replacing:
   a. Log in to the IPMI interface of the node (iDRAC in this case).
   b. Select **Storage** from the menu, and then select **Physical Disks**.
   c. Locate the cache drive.

      Cache drives are of lesser capacity (375 GB) than storage drives, and are PCIe SSDs.

   d. Look for the slot number listed for cache drive.

      This is the physical slot from which you should remove the drive.

3. Now that you have identified the drive, physically remove it as follows:
   a. Identify the drive slot number in the chassis.

      The following image shows the front of the server with the slot numbering shown for each drive:

   b. Press the button on the drive.

      The latch comes out.

   c. Physically pull the drive out from the slot.

      Ensure that you handle drives very carefully.

      After you physically remove the drive, the drive state changes to **Failed** in the Element UI.

4. Make a note of the model number and the ISN (serial number) of the new cache drive.
5. Insert the replacement drive by carefully pushing the drive into the slot using the latch and closing the latch.

   The drive powers on when inserted correctly.

6. Perform the following steps to verify the new drive details in iDRAC:
   a. Select **Maintenance** > **System Event Log**. You see an event logged for the drive that you added.
   b. Select **Storage** from the menu, and then select **Physical Disks**.
   c. Verify that the new drive you inserted shows up in the corresponding slot in the UI.
   d. To find the serial number of the drive, select the **+** icon next to each PCIe SSD.
7. Add the new cache drive information in the `sf_sds_config.yaml` file for the node in which you replaced the drive.

   The `sf_sds_config.yaml` file is stored in `/opt/sf/`. This file includes all the information about the drives in the node. Every time you replace a drive, you should enter the replacement drive information in this file. For more information about this file, see Contents of the sf_sds_config.yaml file.

   a. Establish an SSH connection to the node by using PuTTY.

b. In the PuTTY configuration window, enter node MIP address (that you made a note of from the Element UI earlier) in the **Host Name (or IP address)** field.

c. Select **Open**.

d. In the terminal window that opens, log in with your username and password.

e. Run the `nvme list` command to list the NMVe devices.

You can see the model number and serial number of the new cache drive. See the following sample output:

```
Node             SN                   Model                                    Namespace Usage                      Format           FW Rev
---------------- -------------------- ---------------------------------------- --------- -------------------------- ---------------- --------
/dev/nvme0n1     PHLJ029506A54P0DGN   INTEL SSDPE2KX040T8                      1           4.00  TB /   4.00  TB    512   B +  0 B   VDV10131
/dev/nvme1n1     PHKE91400006375AGN   INTEL SSDPE21K375GA                      1         375.08  GB / 375.08  GB    512   B +  0 B   E2010435
/dev/nvme2n1     PHLJ030004VJ4P0DGN   INTEL SSDPE2KX040T8                      1           4.00  TB /   4.00  TB    512   B +  0 B   VDV10131
/dev/nvme3n1     PHLJ029507NB4P0DGN   INTEL SSDPE2KX040T8                      1           4.00  TB /   4.00  TB    512   B +  0 B   VDV10131
/dev/nvme4n1     PHLJ030004W84P0DGN   INTEL SSDPE2KX040T8                      1           4.00  TB /   4.00  TB    512   B +  0 B   VDV10131
/dev/nvme5n1     PHLJ030101RS4P0DGN   INTEL SSDPE2KX040T8                      1           4.00  TB /   4.00  TB    512   B +  0 B   VDV10131
/dev/nvme6n1     PHLJ029509QX4P0DGN   INTEL SSDPE2KX040T8                      1           4.00  TB /   4.00  TB    512   B +  0 B   VDV10131
/dev/nvme7n1     PHLJ030101S44P0DGN   INTEL SSDPE2KX040T8                      1           4.00  TB /   4.00  TB    512   B +  0 B   VDV10131
/dev/nvme8n1     PHLJ029509QZ4P0DGN   INTEL SSDPE2KX040T8                      1           4.00  TB /   4.00  TB    512   B +  0 B   VDV10131
/dev/nvme9n1     PHLJ030101RW4P0DGN   INTEL SSDPE2KX040T8                      1           4.00  TB /   4.00  TB    512   B +  0 B   VDV10131
```

f. Add the new cache drive information in `/opt/sf/sf_sds_config.yaml`.

You should replace the existing cache drive model number and serial number with the corresponding information for the new cache drive. See the following example:

```
schemaVersion: "2.0"

network:
  managementInterface: "team0"
  storageInterface: "team1"
dataDrives:
  - "/dev/nvme0n1"
  - "/dev/nvme2n1"
  - "/dev/nvme3n1"
  - "/dev/nvme4n1"
  - "/dev/nvme5n1"
  - "/dev/nvme6n1"
  - "/dev/nvme7n1"
  - "/dev/nvme8n1"
  - "/dev/nvme9n1"
cacheDevices:
  - "/dev/disk/by-id/nvme-INTEL_SSDPE21K375GA_PHKE913200Z3375AGN"
```

Model number

Serial number

g. Save the `/opt/sf/sf_sds_config.yaml` file.

8. Perform the steps for the scenario that is applicable to you:

Scenario	Steps
The new inserted cache drive shows up after you run the `nvme list` command	1. Run `# systemctl restart solidfire`. This takes around three minutes. 2. Check the `solidfire` status by running `system status solidfire`. 3. Go to step 9.

Scenario	Steps
The new inserted cache drive does not show up after you run the `nvme list` command	1. Reboot the node. 2. After the node reboots, verify that the `solidfire` services are running by logging in to the node (using PuTTY), and running the `system status solidfire` command. 3. Go to step 9.

> Restarting `solidfire` or rebooting the node causes some cluster faults, which eventually clear in about five minutes.

9. In the Element UI, add back the metadata drive that you removed:
   a. Select **Cluster** > **Drives** > **Available**.
   b. Select the icon under Actions, and select **Add**.
10. Refresh your Element UI after the block sync job completes.

    You can see that the alert about the drive available has cleared along with other cluster faults.

Find more information

- NetApp SolidFire Resources Page
- Documentation for earlier versions of NetApp SolidFire and Element products

Collect container logs

Learn more about the SolidFire eSDS containers and where to get the logs associated with them. The information provided here is to help you get started with collecting logs for troubleshooting. It is best to engage NetApp Support, where trained engineers well versed with log analysis can help resolve issues.

SolidFire eSDS includes the following containers:

- Element container (`element`): Hosts all of the Element services in a single container.
- Element auth container (`element_auth`): Provides multifactor authentication (MFA) and session authentication tokens for the management UIs.
- Network watchdog container (`sfnetwd`): Provides heartbeat monitoring of the local Element instance and failover of the virtual IPs (MVIP and SVIP).

To access the containers on an eSDS node via SSH, the owner of the node should enable SSH and provide the login credentials. After you have SSH access to the host system, you can check the status of one or more containers by using the `podman ps` command. See the example below:

```
# podman ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
f6b8817c024a localhost/solidfire-element:12.2.0.777 --config /sf/etc/...
11 seconds ago Up 10 seconds ago sfnetwd
c3fed6141259 localhost/solidfire-auth:12.2.0.777 --config /sf/etc/... 11
seconds ago Up 11 seconds ago element_auth
1ffa8289c701 localhost/solidfire-element:12.2.0.777 --config /sf/etc/...
12 seconds ago Up 12 seconds ago element
#
```

> If one of the containers is not up or running, do not use Podman to directly control (stop or start) the containers. Element software is controlled via the `solidfire systemd` service unit. This service uses `elementctl` to orchestrate the software running in the three Element containers. Using `systemctl` to control the Solidfire service on the host is the recommended and supported way to start and stop Element software on any node; however, doing so on a live cluster should be only done under the guidance from Support.

All of the logs related to SolidFire eSDS can be found in `/var/log/solidfire/` on the host, which is persisted across container instances. This directory contains the logs from the Element container, and contains the `element_auth/` and `sfnetwd/` subdirectories, which contain the logs from the `element_auth` and `sfnetwd` containers, respectively. From inside a container, you can access container-specific logs at `/var/log`.

Use the NetApp Hybrid Cloud Control UI to collect logs

You can collect logs to send to NetApp Support to help with diagnosing issues with your SolidFire eSDS clusters.

Keep the following considerations in mind for collecting logs:

- Do not use Podman commands to gather logs.
- NetApp Support uses One Collect for host log collection. For an optimal support experience, you should have installed One Collect.

Steps

1. Open the IP address of the management node in a web browser. For example:

   ```
   https://[management node IP address]
   ```

2. Log in to NetApp Hybrid Cloud Control by providing the storage cluster administrator credentials.
3. From the Dashboard, select the menu on the upper right.
4. Select **Collect Logs**.
5. Select a date range in the **Date Range** drop-down menu to specify what dates the logs should include.

   If you specify a custom start date, you can select the date to begin the date range. Logs will be collected from that date up to the present time.

6. In the **Log Collection** section, select the storage cluster or specific storage nodes.
7. Select **Collect Logs** to start log collection.

   Log collection runs in the background, and the page shows the progress.

   > Depending on the logs you collect, the progress bar might remain at a certain percentage for several minutes, or progress very slowly at some points.

8. Select **Download Logs** to download the log package.

   The log package is in a compressed UNIX `.tgz` file format.

Find more information

- NetApp SolidFire Resources Page
- Documentation for earlier versions of NetApp SolidFire and Element products

Links to troubleshooting KB articles

Find the links to the knowledgebase articles for help with troubleshooting your SolidFire eSDS system.

- How to make interface names the same for all the nodes in a cluster (login required)
- How to change the MTU size (login required)
- How to resolve Podman container crash issues (login required)
- How to disable IPv6 for SolidFire eSDS (login required)
- How to change the bonded interfaces on an eSDS cluster while in Production (login required)

This is not a comprehensive list. You can search the Knowledgebase for more articles.

Find more information

- NetApp SolidFire Resources Page
- Documentation for earlier versions of NetApp SolidFire and Element products

Contents of the `sf_sds_config.yaml` file

Every node has a `sf_sds_config.yaml` file, which includes the list of hardware that you specify to be consumed by the SolidFire eSDS services. After you replace a drive, you should add the replacement drive information in this file for the node you replaced the drive from. This file is stored in `/opt/sf/`. This file includes all the information about the drives in the node. You should enter the details of the new drive in this file every time you add a new drive.

Here are the **required** contents of the file:

Key	Default	Type	Description
schemaVersion	"2.0"	String	The schema version number for the file.
network	N/A	List	Allowed values: managementInterface,storageInterface
managementInterface	"team0"	String	The name of the pre-configured, redundant, host OS network interface to use for management and cluster management traffic.
storageInterface	"team1"	String	The name of the pre-configured, redundant, host OS network interface to use for storage network traffic (iSCSI traffic).
dataDrives	N/A	List	The list of host OS paths to physical storage devices being used by SolidFire eSDS. You can specify this as complete paths to the block or NVME devices. The following are supported complete path examples: /dev/disk/by-id/wwn-xxxx-xxxx-EXAMPLE, /dev/disk/by-uuid/nvme-xxxx-xxxxx-EXAMPLE, and /dev/sda1
cacheDevices	N/A	List	The host OS path to the physical device being used as the cache device by SolidFire eSDS. You should specify it as a list entry. The following is an example of a supported complete path: /dev/disk/by-id/nvme-nvme.8086-XXXXXXX-XXXXX-XXXXXX-EXAMPLE

Find more information

- NetApp SolidFire Resources Page
- Documentation for earlier versions of NetApp SolidFire and Element products

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